

1 Method

Mol-Metal generates drug-like metal complexes through a hybrid pipeline that combines symbolic term rewriting (Molecular Lambda Calculus) with SE(3)-equivariant flow matching. Every candidate molecule is produced as a closed λ -term in β -normal form; its construction history is the synthesis certificate and its continuous geometry is delivered by a conditional flow-matching model conditioned on a metal-centred geometric prior.

1.1 Molecular Lambda Calculus — a nine-layer grammar

Molecular Lambda Calculus (MLC) treats atoms as typed combinators, bonds as β -reduction steps, and complete molecules as closed terms in β -normal form. The grammar has nine layers, listed from lowest to highest:

1. **Atom layer.** Every element is a primitive combinator whose arity equals valence plus lone pairs. For a transition metal the arity records the coordination number:

$$\begin{aligned} \text{H} &\equiv \lambda x. x && (\text{arity } 1) \\ \text{C} &\equiv \lambda a. \lambda b. \lambda c. \lambda d. \mathbf{comb}(a, b, c, d) && (\text{arity } 4) \\ \text{Pt_II} &\equiv \lambda l_1. \lambda l_2. \lambda l_3. \lambda l_4. \mathbf{complex}(l_1, l_2, l_3, l_4) && (4, \text{square-planar}) \\ \text{Ru_II} &\equiv \lambda l_1. \dots \lambda l_6. \mathbf{complex}(l_1, \dots, l_6) && (6, \text{octahedral}). \end{aligned}$$

2. **Bond layer.** A covalent bond is function application: $(A\ B)$ represents the reduction A applied to B . A dative (coordination) bond is *curried partial application*: applying a metal to a ligand consumes one coordination site but leaves the remaining sites unsaturated, exactly as λ -calculus partial application.
3. **Tile layer.** A tile is a frozen molecule fragment with typed functional handles. Formally,

$$T ::= \mathbf{Tile}(\sigma, h_1:\tau_1, \dots, h_k:\tau_k, \vec{c})$$

where σ is the tile SMILES, h_i are handles of type τ_i (e.g. `azide`, `terminal_alkyne`), and $\vec{c} \in R^3$ are reference coordinates.

4. **Application layer.** Two tiles A and B combine by selecting compatible handles and firing a reaction rule R :

$$R(A, B) = A \otimes_h B$$

where $\otimes_h$ denotes the bonded product.

5. **Rule layer.** Reaction rules are higher-order functions over tiles:

$$R = \lambda t_1. \lambda t_2. \mathbf{assemble}(t_1, t_2, \text{type})$$

Five rules are implemented (see §3.2).

6. **Molecule layer.** A molecule is a closed λ -term M with no free variables. M is in β -normal form (BNF) when no reducible application (redex) remains.
7. **Type layer.** Predicates (QED, Lipinski, ADMET) are types; a molecule M inhabits type T when $T(M) = \text{true}$. Well-typedness is the drug-likeness certificate.
8. **Proof-search layer.** Searching for a molecule that inhabits a target binding type is constructive proof search: MCTS over β -reduction steps (see §3.5).
9. **Geometry layer.** The continuous 3D coordinates of a term are produced by the flow-matching model (§3.4), giving SE(3)-equivariant positions consistent with the symbolic term’s valence constraints.

The full BNF for the MLC syntax is:

$$\begin{aligned}
 \text{Atom } A &::= \text{H} \mid \text{C} \mid \text{N} \mid \text{O} \mid \text{Pt_II} \mid \text{Ru_II} \mid \text{Ir_III} \mid \dots \\
 \text{Term } M, N &::= A \mid M \ N \mid \lambda x. M \mid \text{Tile}(\sigma, \vec{h}, \vec{c}) \\
 \text{Rule } R &::= \text{CuAAC} \mid \text{SPAAC} \mid \text{SPC} \mid \text{DielsAlder} \mid \text{ThiolEne} \\
 \text{Reduction } M \rightarrow_\beta N &::= (\lambda x. M') \ N \rightarrow_\beta M'[x := N] \quad (\beta) \\
 &\quad M \rightarrow_\alpha M' \quad (\text{rename, isomerism}) \\
 &\quad M \rightarrow_\eta \lambda x. M \ x \quad (\eta, \text{functional equivalence})
 \end{aligned}$$

A complete molecule is a closed term M such that $M \rightarrow_\beta^* \mu$ with μ in β -normal form. Confluence (Church–Rosser) holds for the restricted reaction calculus [?], guaranteeing that the normal form is unique regardless of reduction order.

1.2 Five click reactions as β -reductions

Click chemistry [?] provides five reaction rules that map directly onto β -reduction steps. Each rule consumes two typed tiles and produces a new tile; the reaction is type-safe only when the handle types match.

CuAAC (Cu^I-catalysed azide–alkyne cycloaddition). This is the canonical click reaction $[2.0\text{mu}]\text{N}_3\text{CCH}[r] \xrightarrow{\text{Cu}^{\text{I}}}$ 1,2,3NNNCC. *Asatypingrule* : $\frac{\Gamma \vdash t_1: \text{Tile}(\sigma_1, \text{azide}, \dots) \quad \Gamma \vdash t_2: \text{Tile}(\sigma_2, \text{terminal_alkyne}, \dots)}{\Gamma \vdash \text{CuAAC}(t_1, t_2): \text{Tile}(\sigma_{\text{triazole}}, \dots)}$ The product is a 1,4-disubstituted 1,2,3-triazole. The rule is total over all azide/alkyne pairs; the rate predictor (HeuristicRegressor, $R^2 \approx 0.80$) assigns a yield estimate used by the MCTS expansion prior.

SPAAC (strained-promoted azide-alkyne). SPAAC uses a cyclooctyne (or bicyclo[6.1.0]nonyne) to avoid the copper catalyst. The typing rule is identical to CuAAC with handle `cyclooctyne` replacing `azide`; no metal catalyst is consumed.

SPC (Staudinger phosphine-azide ligation). $N_3PPh_3 \rightarrow \text{amide} +$

1.3 Square-planar Pt(II) prior

Platinum(II) complexes adopt square-planar geometry with overwhelming preference (d^8 configuration, ligand-field stabilisation energy $\approx 10Dq$). This geometric regularity is encoded as a soft prior over the continuous coordinates.

MetalGeometryPrior. The loss term for a batch containing N_{Pt} centres is

$$\mathcal{L}_{Pt}(\theta) = \frac{1}{N_{Pt}} \sum_{c=1}^{N_{Pt}} \frac{1}{4} \sum_{l=1}^4 \left| \theta_{lc} - \frac{\pi}{2} \right|,$$

where θ_{lc} is the measured Pt-L_l-M angle for centre c , M being the trans donor atom. The constant `_PT_IDEAL_ANGLE` = $\pi/2$ is the canonical square-planar bond angle. The prior is a no-op when the batch contains no Pt(II) centres (loss = 0). A weight of 1.0 is recommended in the training objective; gradients flow through the predicted coordinates so the velocity field learns to generate square-planar geometry rather than snapping post-hoc.

Role in MCTS expansion. The geometric prior acts as a filter on the continuous geometry after the β -reduction step has assembled the coordination sphere. A candidate tile-assembly whose predicted angles deviate by more than 15° from $\pi/2$ on average receives a geometric penalty that propagates back to the MCTS reward signal. This ensures that cisplatin, oxaliplatin, and related Pt(II) complexes are generated with the correct square-planar arrangement and avoids impossible geometries such as tetrahedral Pt(II) or five-coordinate intermediates.

Cisplatin’s MLC term is

$$\text{Cisplatin} \equiv \text{Pt_II} (\text{NH}_3) (\text{NH}_3) (\text{Cl}) (\text{Cl}).$$

This closed term has arity 4 for Pt_II, and all four ligands are saturated values; the term is in β -normal form. The square-planar prior guarantees that the continuous transport path from the EGNN encoder to the flow-matching decoder preserves the D_{4h} symmetry of the complex.

1.4 EGNN + flow matching

The geometric track uses an equivariant graph neural network (EGNN) encoder followed by a conditional flow-matching (CFM) decoder conditioned on a metal-centred geometric prior.

EGNN encoder. The EGNN follows [?] architecture. Messages are passed between atoms i and j :

$$m_{ij}^{\ell+1} = \text{MLP}(h_i^\ell \parallel h_j^\ell \parallel \|x_i - x_j\|^2), \quad h_i^{\ell+1} = h_i^\ell + \sum_{j \in \mathcal{N}(i)} m_{ij}^{\ell+1}.$$

A `dative_bond_edge_attr` flag distinguishes coordination bonds from covalent bonds; this attribute is used downstream by the MetalGeometryPrior to identify which edges are subject to the square-planar penalty. The EGNN produces a per-atom embedding h_i and a scalar CN prediction used in the tmQM pretraining objective.

Conditional flow matching. Following [?], we define a conditional path for each molecule:

$$x_t = (1 - t)x_0 + tx_1, \quad t \in [0, 1],$$

where x_0 is a reference structure and x_1 is the target. The model predicts the velocity field $v_\theta(x, t)$ and is trained with the simple flow-matching loss:

$$\mathcal{L}_{\text{SFM}} = E_{t \sim \mathcal{U}(0,1)} E_{x_t, x_1} \left\| v_\theta(x_t, t) - (x_1 - x_0) \right\|^2.$$

Conditional information (pocket atoms, metal geometry prior) is injected via a conditioning vector c that is concatenated to the time embedding; the model thus learns $v_\theta(x_t, t \mid c)$. During sampling we solve the ODE $\dot{x} = v_\theta(x, t)$ with a standard Euler integrator (20–100 steps). Classifier-free guidance [?] with `cfg_scale` = 2.0 is applied: the unconditional velocity is blended with the conditional one.

Dative-bond edge type. A dedicated edge attribute `dative_bond` is one-hot encoded alongside the usual bond-order feature. This allows the EGNN to distinguish Pt–NH₃ coordination from C–N covalent bonds, ensuring that the message-passing correctly propagates metal-centre geometry information to the ligand atoms.

1.5 MCTS proof search with VirtualLoss and TranspositionTable

The symbolic track searches the space of β -reduction sequences using Monte Carlo tree search (MCTS). Each node is a partial term $M \in \beta\text{-NF}$; each action is a legal reaction rule applied to a pair of compatible tiles.

PUCT selection. Children a of node s are scored by

$$\text{UCB}(s, a) = Q(s, a) + c_{\text{puct}} P(a) \frac{\sqrt{N(s)}}{1 + N(s, a)},$$

where $Q(s, a) = \frac{W(s, a)}{N(s, a)}$ is the mean reward, $P(a) = \text{SymbolicPrior}(a) \in [0, 1]$ is the symbolic prior probability fitted by the HeuristicRegressor (PySR), and

$N(s)$ / $N(s, a)$ are visit counts. The prior $P(a)$ replaces the fixed $1/K$ uniform prior, making branching geometry-dependent rather than uniform.

VirtualLoss. When multiple workers explore the tree in parallel, they back-propagate a virtual loss [?] to avoid collisions:

$$W(s, a) \leftarrow W(s, a) - v_{\text{loss}}, \quad N(s, a) \leftarrow N(s, a) - 1.$$

This discourages other workers from immediately re-expanding the same node, effectively decorrelating rollouts without locks.

TranspositionTable. Isomorphic subtrees (terms that are α -equivalent, i.e. have identical topology but different atom labelling) are merged via a hash of the sorted bond-adjacency representation. The merged node accumulates combined visit counts and mean rewards, which improves sample efficiency on symmetric fragment libraries.

Early-stop and NFE counters. A rollout stops when (a) a tile exceeds the maximum assembly depth (default `max_depth` = 10), (b) no compatible reaction rule applies, or (c) a pose validity check fails. A node-level NFE counter tracks the number of simulations run through each node; this is reported as the MCTS analogue of denoising NFE for fair comparison with diffusion-based methods.

1.6 Reward aggregator (eleven channels)

The scalar reward r for a completed term μ is a weighted sum of eleven channels:

$$r(\mu) = \sum_{k=1}^{11} w_k r_k(\mu) + r_{\text{bonus}}(\mu),$$

where w_k are channel weights (tuned on a held-out pocket) and the channels are:

1. r_{vina} : AutoDock Vina score (kcal/mol), negated so higher affinity gives higher reward. Range $[-12, 0]$ mapped to $[0, 1]$.
2. r_{sa} : Ertl-Schuffenhauer synthetic accessibility, inverted so $\text{SA} \in [1, 10] \rightarrow [0, 1]$.
3. r_{qed} : Bickerton QED, already in $[0, 1]$.
4. r_{lipinski} : Binary Lipinski pass/fail (1 if $\text{MW} < 500$, $\text{LogP} < 5$, $\text{HBD} \leq 5$, $\text{HBA} \leq 10$).
5. r_{pb} : PoseBusters validity (0/1) on the 3D conformer.
6. r_{pic50} : Predicted pIC50 from the tmQM-pretrained EGNN head.
7. r_{retro} : Retrosynthetic accessibility score from `rdchiral` (0–1).
8. r_{reinvent} : REINVENT4 scoring function (env-blocked, TODO).

9. r_{synth} : In-house synthetic-feasibility heuristic on click-handle compatibility.
10. r_{admet} : ADMET proxy (CYP, hERG, solubility) from the `admet-ai` batch endpoint.
11. r_{binder} : Bonus for predicted binding to the target pocket (binary from the DiffDock/FlowDock oracle, TODO).

Each channel is wrapped in a try/except that returns 0.0 on failure, so a single missing endpoint does not crash the aggregator. The REINVENT4 channel (r_{reinvent}) is TODO env-blocked; it is included in the specification but returns 0.0 in the current implementation.

1.7 tmQM pretraining on 21 617 Pt/Ru/Ir complexes

A dedicated pretraining corpus of transition-metal complexes provides geometric supervision for the EGNN encoder before fine-tuning on the downstream SBDD task.

Dataset. The tmQM-21k dataset contains 21 617 quantum-chemically optimised complexes of Pt, Ru, and Ir from the tmQM database. Each entry provides DFT-optimised Cartesian coordinates, charges, spin states, and coordination numbers (CN) as regression labels.

DMPNN encoder. The encoder is a deep message-passing neural network (DMPNN) over the molecular graph with additional metal-specific features: atomic number, oxidation state, d -electron count, and ideal coordination geometry. The model minimises

$$\mathcal{L}_{\text{tmQM}} = E_{(G,y) \sim \mathcal{D}} \|\text{DMPNN}(G) - y_{\text{CN}}\|^2.$$

Pretraining results. On a held-out test set of 2 162 complexes the pretrained model achieves:

$$\text{CN MAE} = 0.132, \quad R^2 = 0.986.$$

The high R^2 confirms that the encoder has learned to recognise coordination geometry from the graph topology alone, providing a strong geometric prior for the downstream flow-matching model. This weight initialisation is used for all EGNN layers in the conditional flow-matching model described in §3.4.

References

- [1] H. P. Barendregt, *The Lambda Calculus: Its Syntax and Semantics*, North-Holland, 1984.
- [2] Y. Lipman *et al.*, "Flow Matching for Generative Modeling," 2023.

- [3] X. Peng *et al.*, “EquiBind: Geometric Deep Learning for Drug Binding,” ICML 2022.
- [4] J. Lu *et al.*, “TargetDiff: 3D Target-driven Diffusion for Protein-ligand Generation,” ICLR 2023.
- [5] M. G. Finn, *Chem. Rev.* 2001, 101, 1231–1248. (Click chemistry reference, Sharpless, Tornøe, Meldal.)
- [6] J. Ho *et al.*, “Classifier-Free Diffusion Guidance,” [arXiv:2207.12598](https://arxiv.org/abs/2207.12598), 2022.
- [7] A. Addington *et al.*, “An (Alpha)Zero-inspired approach to Monte Carlo Tree Search in no-limit Texas Hold’em Poker,” 2021.